

SREEHARI P S

AI Engineer | Generative AI | Machine Learning | Deep Learning

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PROFESSIONAL SUMMARY

AI Engineer with 6+ months of hands-on experience building intelligent applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) pipelines, and conversational AI systems. Proficient in deep learning frameworks (PyTorch, TensorFlow), NLP, prompt engineering, LLM fine-tuning, and deploying production-grade ML systems integrated with cloud APIs. Experienced with agentic AI orchestration, computer vision pipelines, and edge AI deployment.

TECHNICAL SKILLS

GenAI & LLMs: LangChain, RAG Pipelines, Agentic AI, LLM Fine-tuning, Prompt Engineering, Ollama (Llama 3), ChromaDB, Intent Recognition

ML / Deep Learning: PyTorch, TensorFlow, ONNX, Scikit-learn, Model Pruning, Ensemble Methods

Computer Vision: YOLO (v12n), OpenPifPaf, OpenCV, Pose Estimation, Thermal Imaging, RoboFlow

Dev & Cloud: FastAPI, Flask, Streamlit, REST APIs, AWS, Docker, CI/CD, Git

Data & Tools: Python, SQL, NumPy, Pandas, EDA, Feature Engineering, ETL Pipelines, Power BI, NVIDIA Jetson Nano

WORK EXPERIENCE

AI Engineer Intern | IHUB Robotic

Jan 2026 – Mar 2026

- Engineered an end-to-end RAG pipeline using LangChain and ChromaDB with FastAPI endpoints for enterprise document retrieval, reducing query resolution time by 40% vs. baseline keyword search.
- Deployed a YOLO-based thermal fire detection model on NVIDIA Jetson Nano achieving 30 FPS edge inference, integrated into an autonomous ROS robot with automated alert transmission.
- Delivered 5+ production NLP and CV pipelines covering intent recognition, object detection, and tokenization, cutting deployment cycle time by 60% via standardized pipeline templates.

Machine Learning Intern | Prodigal AI

Mar 2025 – Jun 2025

- Optimized YOLOv12n models on a 15,000+ image dataset via pruning and ensemble learning, reducing inference latency by 25% while maintaining production accuracy under CI/CD benchmarks.
- Implemented OpenPifPaf pose estimation across 17 body keypoints, improving CV accuracy by 18% for downstream health-tech applications.

Data Engineering Intern | Amagine

Oct 2024 – Dec 2024

- Built ETL pipelines processing 10M+ records with EDA and JSON-based feature transformations, improving ML data quality by 35%; SQL and API optimizations reduced pipeline bottlenecks by 50%.

EDUCATION

B.E. – Artificial Intelligence & Machine Learning

Sep 2021 – May 2025

Yenepoya Institute of Technology, Mangalore, India | GPA: 8.2 / 10.0

KEY PROJECTS

DocuMind AI | Agentic RAG · LangChain · ChromaDB · FastAPI

Jan 2025 – Present

- Architected a fully local MCP-style agentic RAG system using LangChain ReAct + ChromaDB + FastAPI, achieving 92%+ semantic retrieval accuracy across 50+ documents with zero cloud dependency and enterprise-grade privacy.

Thermal Fire Detection Robot | YOLO · ROS · NVIDIA Jetson Nano

Feb 2026 – Present

- Trained YOLO on 2,000+ custom thermal images achieving mAP@0.5 > 90%; deployed on Jetson Nano at 30 FPS with ROS-integrated autonomous fire-source navigation and real-time alert system.

Exercise Monitoring System | OpenPifPaf · Pose Estimation · Computer Vision

Mar – May 2025

- Built a real-time pose estimation system using OpenPifPaf (17 keypoints) achieving 95%+ form accuracy validated on 500+ video samples from 20+ users for health-tech injury prevention.

ASL Sign Language Translator | TensorFlow · OpenCV · Tkinter

Oct – Dec 2024

- Built a TensorFlow and OpenCV system translating 26 ASL gestures to text and audio with 94% accuracy; deployed offline via Tkinter GUI with pytsx3, ensuring full accessibility with zero external dependencies.